

CSCN8020 Assignment 3

Deep Q-Network Control of the Unitree G1 Left Elbow

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Course: CSCN8020-Reinforcement Learning

1. Video Demonstration GoogleDrive Link (2-3 Minutes):

https://drive.google.com/file/d/1Ozw8JLJhtj1zUTw1kjYm4g_ARNWx5LQ/view?usp=drive_link

(Demonstrates greedy policy evaluation on selected_dqn.pt with epsilon = 0.0 across benchmark target angles).

2. GitHub Repository Link:

https://github.com/caatat741213/CSCN8020_Assignment-3.git

3. Project Summary

This project implements a Deep Q-Network (DQN) agent using PyTorch to control the left elbow joint of the Unitree G1 humanoid robot in MuJoCo. Building on the G1 Primer Workshop, the model learns a discrete action policy to reach target angles between -0.8 and $+0.8$ rad. Evaluated across five hyperparameter configurations, the selected model (Config B) achieved a 100% success rate over 20 benchmark test episodes. Compared to a rule-based baseline, the DQN agent reduced the mean episode length from 24.0 to 19.5 steps and improved angular accuracy by using joint velocity to brake dynamically near target goals.